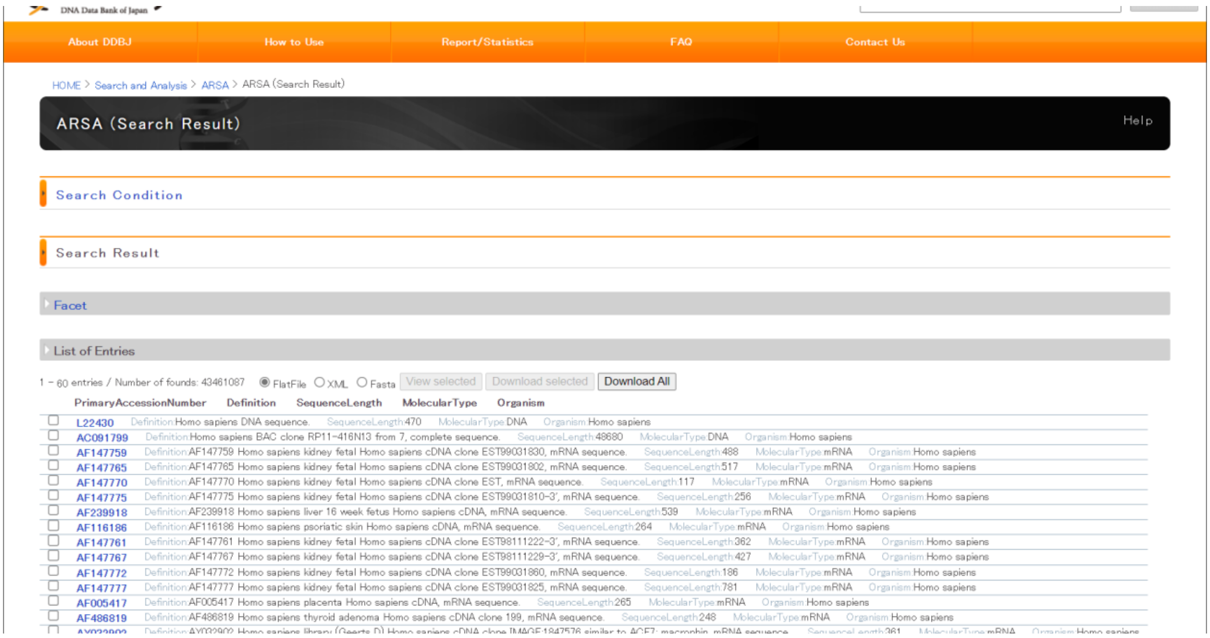
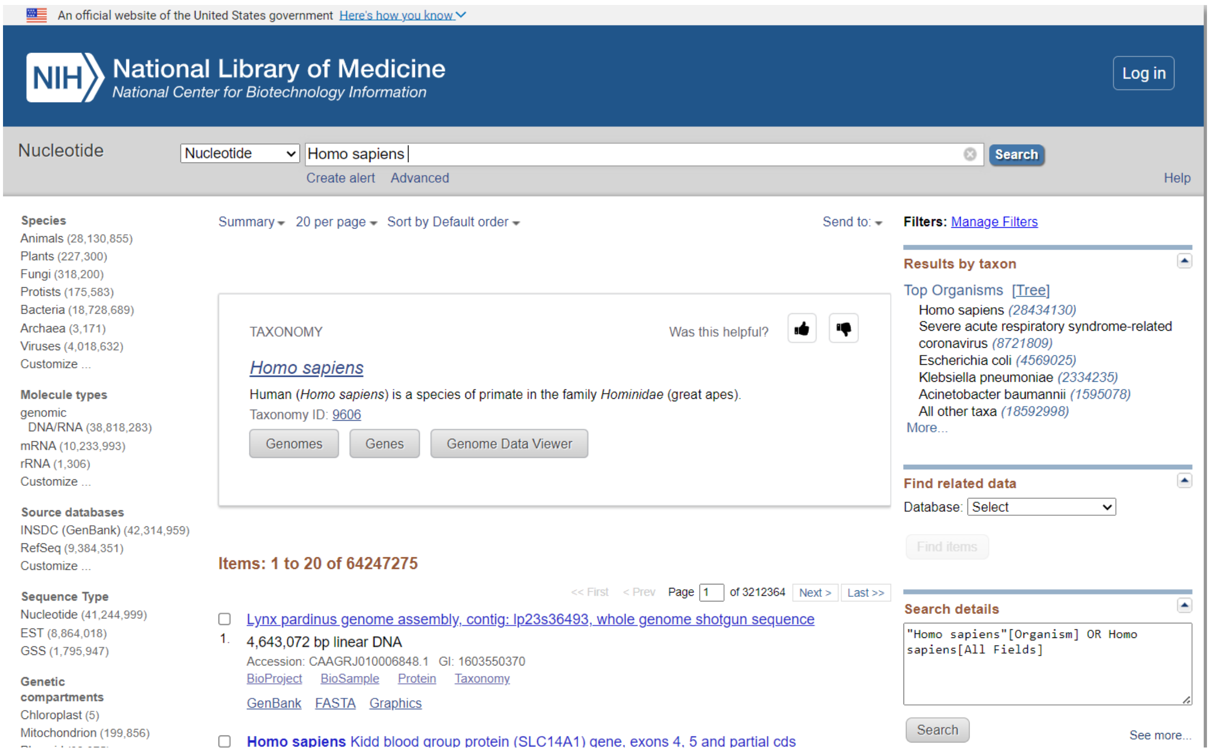
# Bioinformatics: Practical 2

Name: Sidharthan S C

Roll no: BE21B039

1)



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## Text Search

Uses EBI Search to perform a free text search across ENA data. For more detailed usage please refer to the help & documentation section.

**Search term:** Homo sapiens **Search**

Search results for Homo sapiens

- **Assembly**
  - Assembly (1,821)
- **Sequence**
  - Sequence (40,620,498)
  - Sequence (CON) (565,646)
  - Sequence (Standard) (40,054,852)
- **Contig set**
  - Genome assembly contig set (338,633)
  - Transcriptome assembly contig set (9)
  - Targeted locus study contig set (582)
- **Coding**
  - Coding (TSA) (732)
  - Coding (TLS) (2,599,382)
- **Non-coding**
  - Non-coding (43,459,534)
- **Read**

**Assembly** View all 1,821 results.

GCA_002884475.1 YRI_HSA_v1 assembly for Homo sapiens

**Sequence** View all 40,620,498 results.

KJ946236 Homo sapiens Kidd blood group protein (SLC14A1) gene, exons 4, 5 and partial cds.

**Sequence (CON)** View all 565,646 results.

KN707688 Homo sapiens unplaced genomic scaffold decoy00122.

**Sequence (Standard)** View all 40,054,852 results.

KJ946236 Homo sapiens Kidd blood group protein (SLC14A1) gene, exons 4, 5 and partial cds.

**Genome assembly contig set** View all 338,633 results.

CZVB02000000 Homo sapiens genome assembly

**Transcriptome assembly contig set** View all 9 results.

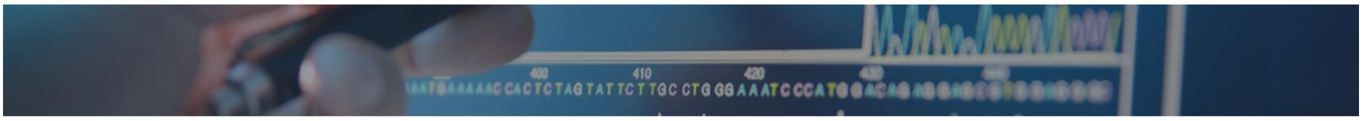
GHIQ01000000 Homo sapiens, TSA project GHIQ01000000 data



DDBJ has 43461087 entries which is less than GenBank: 64247275 while sequences deposited in EMBL is 40620 498.

2)

GC content is 45.681818%



**Your input seq is:**

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Nucleotide Content:

| | Nucleotide content in % |
|---|---|
| AT_content | 52.045455 |
| Adenine_content | 29.318182 |
| Cytosine_content | 16.363636 |
| GC_content | 45.681818 |
| Guanine_content | 29.318182 |
| Keto_GT_content | 52.045455 |
| Purine_AG_content | 58.636364 |
| Thymine_content | 22.727273 |

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3) GEN Bank

GenBank ▾ Send to: ▾

## E. coli lac operator DNA fragment 5' to Z gene

GenBank: X03527.1

FASTA Graphics

Go to:

```
LOCUS       X03527                    55 bp    DNA     linear   BCT 23-OCT-2008
DEFINITION  E. coli lac operator DNA fragment 5' to Z gene.
ACCESSION   X03527 M12479
VERSION     X03527.1
KEYWORDS    operator; protein binding site.
SOURCE      Escherichia coli
  ORGANISM  Escherichia coli
            Bacteria; Pseudomonadota; Gammaproteobacteria; Enterobacterales;
            Enterobacteriaceae; Escherichia.
REFERENCE   1  (bases 1 to 55)
  AUTHORS   Manly,S.P. and Matthews,K.S.
  TITLE     lac operator DNA modification in the presence of proteolytic
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  JOURNAL   J. Mol. Biol. 179 (3), 315-333 (1984)
   PUBMED   6392562
COMMENT     On Mar 11, 2005 this sequence version replaced M12479.1.
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ORIGIN
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Change region shown

Customize view

**Analyze this sequence**

- Run BLAST
- Pick Primers
- Highlight Sequence Features
- Find in this Sequence

**Related information**

- PubMed
- Taxonomy

**Recent activity**

Turn Off Clear

- E. coli lac operator DNA fragment 5' to Z gene — Nucleotide
- Synthetic construct human lysozyme mRNA, complete cds — Nucleotide
- Homo sapiens (64247275) — Nucleotide
- GCF_000151735.1[assembly] NOT nuccore_comp_nuccore[filter] (3143 — Nucleotide
- txid9606[Organism:exp] (1) — Genome

See more...

DDBJ

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LOCUS       X03527                    55 bp    DNA     linear   BCT 23-OCT-2008
DEFINITION  E. coli lac operator DNA fragment 5' to Z gene.
ACCESSION   X03527 M12479
VERSION     X03527.1
KEYWORDS    operator; protein binding site.
SOURCE      Escherichia coli
  ORGANISM  Escherichia coli
            Bacteria; Proteobacteria; Gammaproteobacteria; Enterobacterales;
            Enterobacteriaceae; Escherichia.
REFERENCE   1  (bases 1 to 55)
  AUTHORS   Manly S.P., Shive M.K.
  TITLE     lac operator DNA modification in the presence of proteolytic
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  JOURNAL   J. Mol. Biol. 179(3), 315-333(1984).
   PUBMED   6392562
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EMBL

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ID   X03527; SV 1; linear; genomic DNA; STD; PRO; 55 BP.
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AC   X03527; M12479;
XX
DT   02-JUL-1986 (Rel. 09, Created)
DT   23-OCT-2008 (Rel. 97, Last updated, Version 5)
XX
DE   E. coli lac operator DNA fragment 5' to Z gene
XX
KW   operator; protein binding site.
XX
OS   Escherichia coli
OC   Bacteria; Proteobacteria; Gammaproteobacteria; Enterobacterales;
OC   Enterobacteriaceae; Escherichia.
XX
RN   [1]
RP   1-55
RX   DOI; 10.1016/0022-2836(84)90068-8.
RX   PUBMED; 6392562.
RA   Manly S.P., Shive M.K.;
RT   "lac operator DNA modification in the presence of proteolytic fragments of
RT   the repressor protein";
RL   J. Mol. Biol. 179(3):315-333(1984).
XX
DR   MD5; 5dc87667dc65a72a21939dded7e3a887.
XX
FH   Key             Location/Qualifiers
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FT   variation       36..36
FT                   /note="C is T in lac O-c"
XX
SQ   Sequence 55 BP; 16 A; 9 C; 16 G; 14 T; 0 other;
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//
```

9)

# EC Explorer

[browse EC tree] - [search]

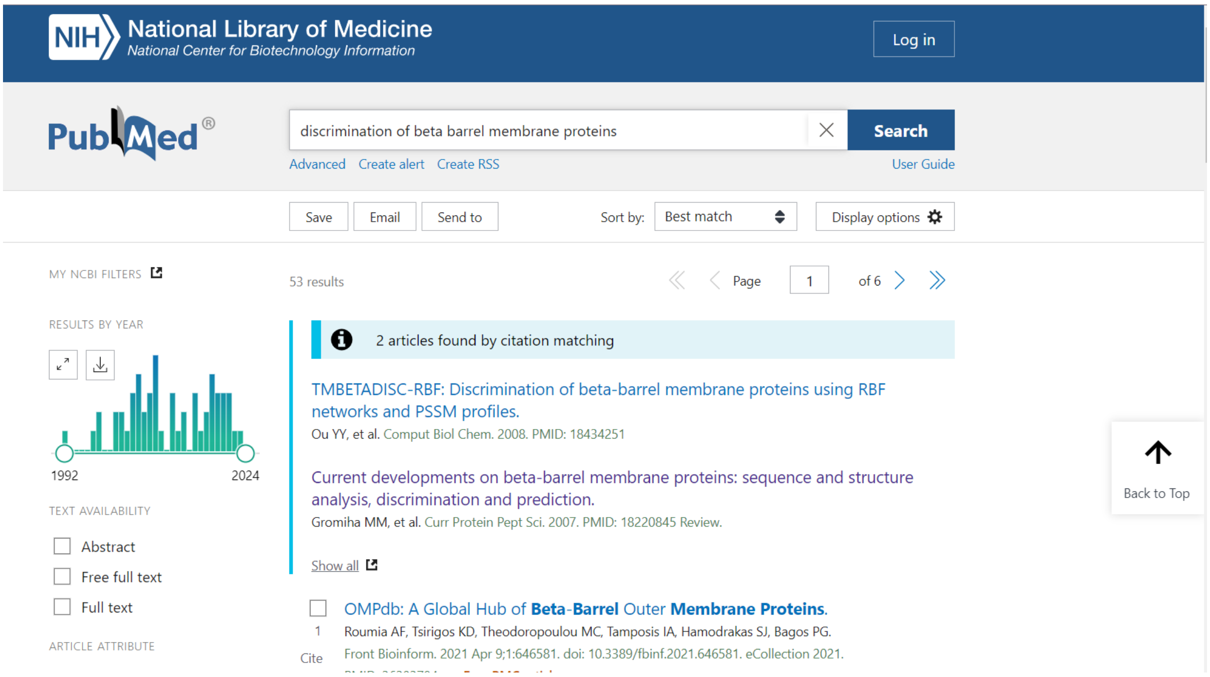
**EC 1.7.2.3 Details**

| |
|---|
| show Brenda entry |
| EC number |
| 1.7.2.3 |
| Accepted name |
| trimethylamine-*N*-oxide reductase |
| Reaction |
| trimethylamine + **2** (ferricytochrome *c*)-subunit + $H_2O$ = trimethylamine *N*-oxide + **2** (ferrocytochrome *c*)-subunit + **2** $H^+$ |
| Other name(s) |
| TMAO reductase, TOR, *torA* (gene name), *torZ* (gene name), *bisZ* (gene name), trimethylamine-*N*-oxide reductase (cytochrome *c*) |
| Systematic name |
| trimethylamine:cytochrome *c* oxidoreductase |
| CAS registry number |
| 37256-34-1 |
| Comment |
| Contains bis(molybdopterin guanine dinucleotide)molybdenum cofactor. The reductant is a membrane-bound multiheme cytochrome *c*. Also reduces dimethyl sulfoxide to dimethyl sulfide. |
| History |
| created 2002, modified 2018 |

**EC Tree**

- 1 Oxidoreductases
  - 1.1 Acting on the CH-OH group of donors
  - 1.2 Acting on the aldehyde or oxo group of donors
  - 1.3 Acting on the CH-CH group of donors
  - 1.4 Acting on the CH-$NH_2$ group of donors
  - 1.5 Acting on the CH-NH group of donors
  - 1.6 Acting on NADH or NADPH
  - 1.7 Acting on other nitrogenous compounds as donors
    - 1.7.1 With $NAD^+$ or $NADP^+$ as acceptor
    - 1.7.2 With a cytochrome as acceptor
      - 1.7.2.1 nitrite reductase (NO-forming)
      - 1.7.2.2 nitrite reductase (cytochrome; ammonia-forming)
      - 1.7.2.3 trimethylamine-*N*-oxide reductase
      - 1.7.2.4 nitrous-oxide reductase
      - 1.7.2.5 nitric oxide reductase (cytochrome *c*)
      - 1.7.2.6 hydroxylamine dehydrogenase
      - 1.7.2.7 hydrazine synthase
      - 1.7.2.8 hydrazine dehydrogenase
      - 1.7.2.9 hydroxylamine oxidase
    - 1.7.3 With oxygen as acceptor
    - 1.7.5 With a quinone or similar compound as acceptor
    - 1.7.6 With a nitrogenous group as acceptor
    - 1.7.7 With an iron-sulfur protein as acceptor
    - 1.7.9 With a copper protein as acceptor
    - 1.7.99 With unknown physiological acceptors
  - 1.8 Acting on a sulfur group of donors
  - 1.9 Acting on a heme group of donors
  - 1.10 Acting on diphenols and related substances as donors
  - 1.11 Acting on a peroxide as acceptor

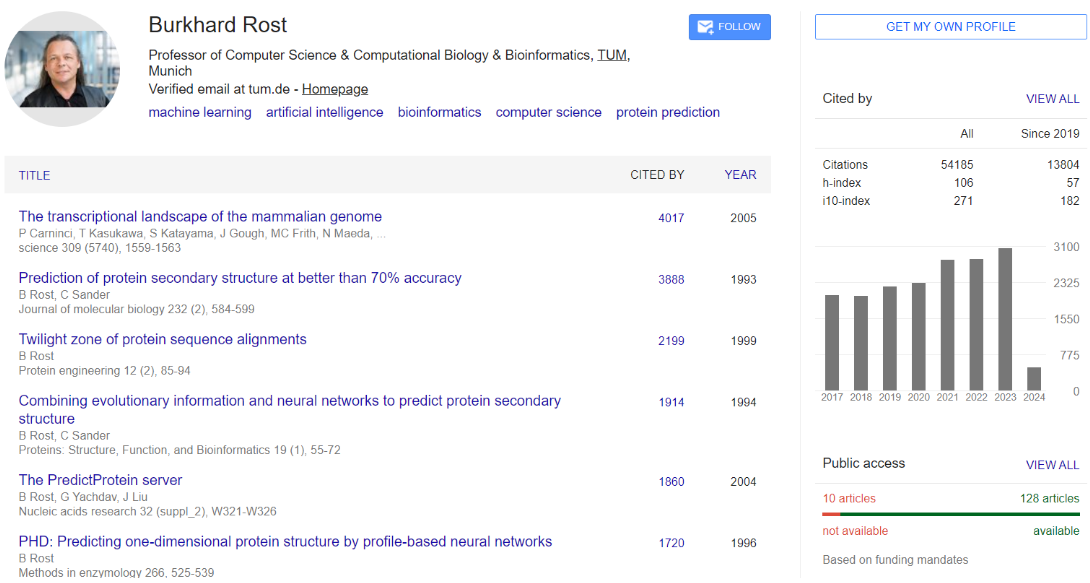
**4)**




| | A | B | C | D | E | F | G | H | I | J | K | L | M | N |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | PMID | Title | Authors | Citation | First Autho | Journal/Bc | Publicatior | Create Dat | PMCID | NIHMS ID | DOI | | | |
| 2 | 8805593 | Human dU | Mol CD, Ha | Structure. | Mol CD | Structure | 1996 | ######## | | | 10.1016/s0969-2126(96)00114-1 | | | |
| 3 | 17557332 | A consensu | Garrow AC | Proteins. 2 | Garrow AC | Proteins | 2007 | ######## | | | 10.1002/prot.21439 | | | |
| 4 | 10210191 | Structural | Juers DH, H | Protein Sci | Juers DH | Protein Sci | 1999 | ######## | PMC2144101 | | 10.1110/ps.8.1.122 | | | |
| 5 | 37432995 | General fe | Montezano | Proc Natl A | Montezano | Proc Natl A | 2023 | ######## | PMC10629564 | | 10.1073/pnas.2220762120 | | | |
| 6 | 19380741 | Single-nucl | Stoddart D | Proc Natl A | Stoddart D | Proc Natl A | 2009 | ######## | PMC2683137 | | 10.1073/pnas.0901054106 | | | |
| 7 | 1565623 | Human au | Marchalor | Proc Natl A | Marchalor | Proc Natl A | 1992 | ######## | PMC48859 | | 10.1073/pnas.89.8.3325 | | | |
| 8 | 30153309 | Select gp1 | Lertjuthap | PLoS Patho | Lertjuthap | PLoS Patho | 2018 | ######## | PMC6130882 | | 10.1371/journal.ppat.1007278 | | | |
| 9 | 22073138 | Protein do | Prakash A, | PLoS One. | Prakash A | PLoS One | 2011 | ######## | PMC3206015 | | 10.1371/journal.pone.0025570 | | | |
| 10 | 33137176 | Novel ribo | Zatopek KI | Nucleic Ac | Zatopek KI | Nucleic Ac | 2020 | ######## | PMC7708050 | | 10.1093/nar/gkaa986 | | | |
| 11 | 20601684 | Structure c | Nureki O, ( | Nucleic Ac | Nureki O | Nucleic Ac | 2010 | ######## | PMC2978374 | | 10.1093/nar/gkq605 | | | |
| 12 | | | | | | | | | | | | | | |

**8)h-index=106**

**citations= 54165**




**5)**

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Kihara D | Search

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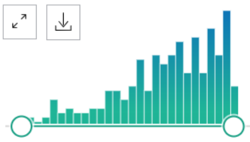
219 results   Page 1 of 22

RESULTS BY YEAR




TEXT AVAILABILITY

- Abstract
- Free full text
- Full text

ARTICLE ATTRIBUTE

1 [Computational protein function predictions.](#)
**Kihara D.**
Methods. 2016 Jan 15;93:1-2. doi: 10.1016/j.ymeth.2016.01.001.
PMID: 26778120   No abstract available.
Cite Share

2 [Computational prediction of disordered binding regions.](#)
Basu S, **Kihara D**, Kurgan L.
Comput Struct Biotechnol J. 2023 Feb 10;21:1487-1497. doi: 10.1016/j.csbj.2023.02.018. eCollection 2023.
PMID: 36851914   Free PMC article.   Review.
Cite Share

3 [Editorial: Expert opinions in protein bioinformatics: 2022.](#)
**Kihara D.**
Front Bioinform. 2024 Jan 5;3:1338560. doi: 10.3389/fbinf.2023.1338560. eCollection 2023.
Cite

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | PMID | Title | Authors | Citation | First Autho | Journal/Bc | Publicatior | Create Dat | PMCID | NIHMS ID | DOI | | | |
| 2 | 11802444 | [Genome ( | Kihara D. | Tanpakush | Kihara D | Tanpakush | 2001 | ######## | | | | | | |
| 3 | 9455244 | [Internet r | Kihara D, K | Tanpakush | Kihara D | Tanpakush | 1997 | ######## | | | | | | |
| 4 | 28262392 | Variability | Monroe L, | Structure. | Monroe L | Structure | 2017 | ######## | PMC53821 | NIHMS854 | 10.1016/j.str.2017.02.004 | | | |
| 5 | 32064000 | 2DKD: a to | DeVille JS, | Source Co | DeVille JS | Source Co | 2020 | ######## | PMC7011505 | | 10.1186/s13029-020-0077-1 | | | |
| 6 | 31698087 | Current pr | Karuppasa | Semin Can | Karuppasa | Semin Can | 2021 | ######## | PMC92335 | NIHMS181 | 10.1016/j.semcancer.2019.10.019 | | | |
| 7 | 25554786 | Structure a | Liu Y, Shen | Science. 2( | Liu Y | Science | 2015 | ######## | PMC43077 | NIHMS656 | 10.1126/science.1261962 | | | |
| 8 | 18535227 | The emerg | Hu JC, Arai | Science. 2( | Hu JC | Science | 2008 | ######## | | | 10.1126/science.320.5881.1289b | | | |
| 9 | 33828153 | Analyzing ( | Jain A, Ter | Sci Rep. 2( | Jain A | Sci Rep | 2021 | ######## | PMC8027171 | | 10.1038/s41598-021-87204-z | | | |
| 10 | 31863054 | A prospect | Chiba S, Ol | Sci Rep. 2( | Chiba S | Sci Rep | 2019 | ######## | PMC6925144 | | 10.1038/s41598-019-55069-y | | | |
| 11 | 31784686 | NNTox: Ge | Jain A, Kiha | Sci Rep. 2( | Jain A | Sci Rep | 2019 | ######## | PMC6884647 | | 10.1038/s41598-019-54405-6 | | | |
| 12 | | | | | | | | | | | | | | |

**6)**

## Similar articles

[Evolution. Tinkering inside the organelle.](#)
Alcock F, Clements A, Webb C, Lithgow T.
Science. 2010 Feb 5;327(5966):649-50. doi: 10.1126/science.1182129.
PMID: 20133559   No abstract available.

[Transport proteins (carriers) of mitochondria.](#)
Wohlrab H.
IUBMB Life. 2009 Jan;61(1):40-6. doi: 10.1002/iub.139.
PMID: 18816452   Review.

[Systematic analysis of the twin cx(9)c protein family.](#)
Longen S, Bien M, Bihlmaier K, Kloeppel C, Kauff F, Hammermeister M, Westermann B, Herrmann JM, Riemer J.
J Mol Biol. 2009 Oct 23;393(2):356-68. doi: 10.1016/j.jmb.2009.08.041. Epub 2009 Aug 21.
PMID: 19703468

[Mitochondrial permeability transition pore opening as a promising therapeutic target in cardiac diseases.](#)
Javadov S, Karmazyn M, Escobales N.
J Pharmacol Exp Ther. 2009 Sep;330(3):670-8. doi: 10.1124/jpet.109.153213. Epub 2009 Jun 9.
PMID: 19509316

[A CaPful of mechanisms regulating the mitochondrial permeability transition.](#)
Di Lisa F, Bernardi P.
J Mol Cell Cardiol. 2009 Jun;46(6):775-80. doi: 10.1016/j.yjmcc.2009.03.006. Epub 2009 Mar 19.
PMID: 19303419   Review.

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11)

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Eukaryota / Metazoa / Chordata / Mammalia / Primates / Hominidae / Homo

## *Homo sapiens* ☆

Human (*Homo sapiens*) is a species of primate in the family *Hominidae* (great apes).

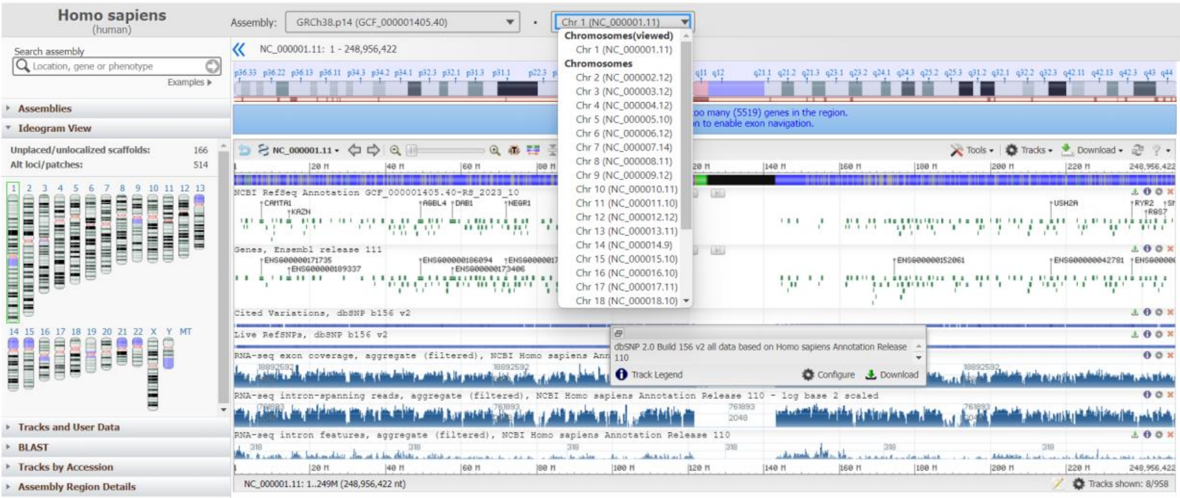
Browse taxonomy

| | |
|---|---|
| Current scientific name | *Homo sapiens* |
| Common name | human |
| Taxonomic rank | species |
| NCBI Taxonomy ID | 9606 |

For more details see NCBI Taxonomy

Feedback

## Genome Data Viewer




NCBI Datasets | Taxonomy | Genome | Gene | Command-line tools | Documentation

Eukaryota / Metazoa / Chordata / Mammalia / Carnivora / Felidae / Felis

## *Felis catus* ☆

Domestic cat (*Felis catus*) is a species of carnivore in the family *Felidae* (cat family).

Browse taxonomy

| | |
|---|---|
| Current scientific name | *Felis catus* |
| Common name | domestic cat, cat, cats |
| Taxonomic rank | species |
| NCBI Taxonomy ID | 9685 |

For more details see NCBI Taxonomy

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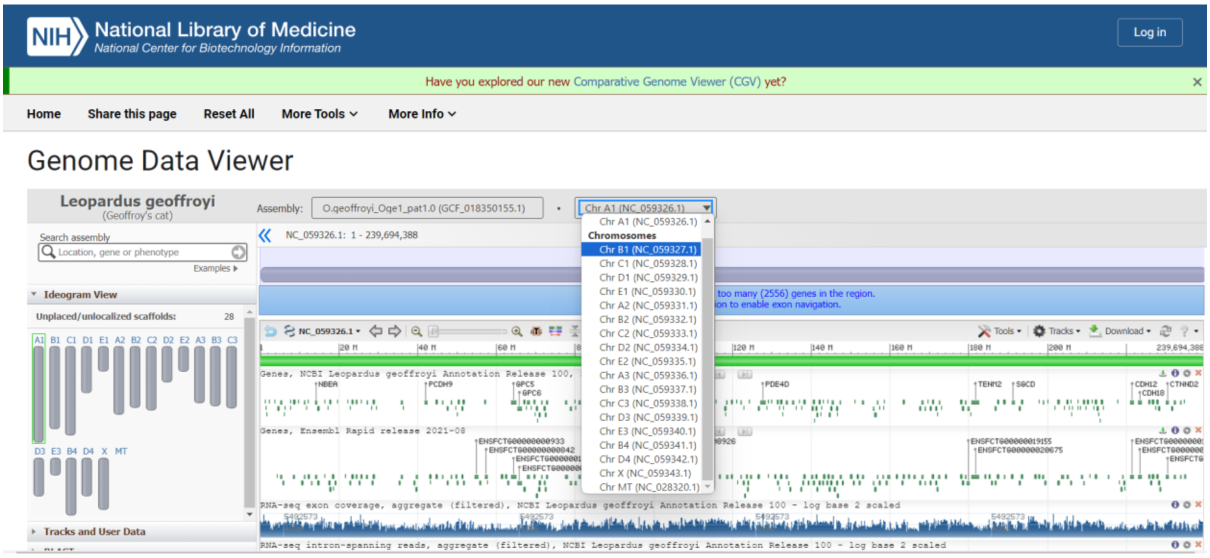
Image may not have been verified for accuracy by NCBI Taxonomy.

External links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Genome Data Viewer
Leopardus geoffroyi
(Geoffroy's cat)
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Chr A1 (NC_059326.1)
NC_059326.1: 1 - 239,694,388
Search assembly
Location, gene or phenotype
Examples
Ideogram View
Unplaced/unlocalized scaffolds:
28
Chromosomes
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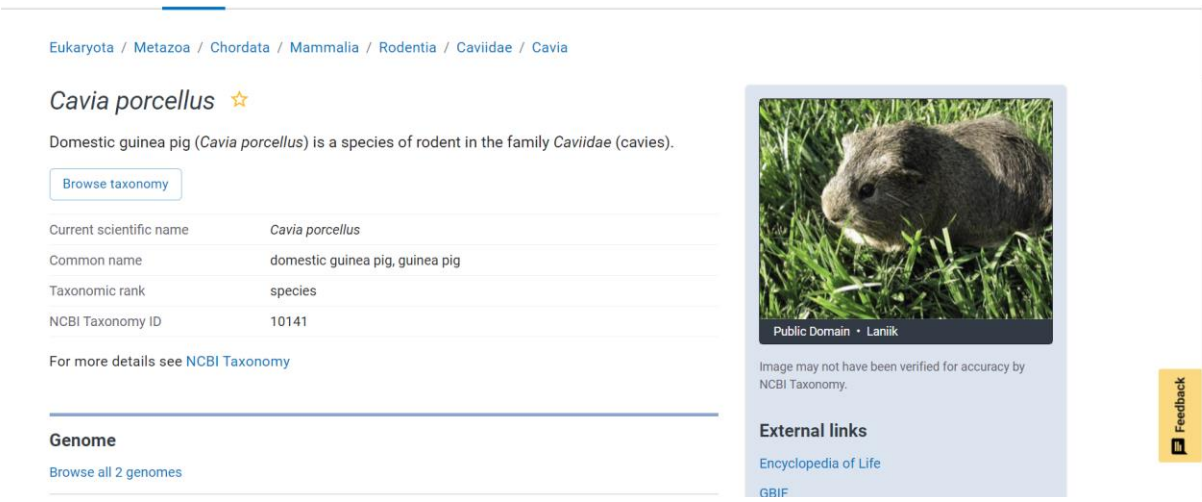
Chr MT (NC_028320.1)
Tracks and User Data
Genes, NCBI Leopardus geoffroyi Annotation Release 100,
Genes, Ensembl Rapid release 2021-08
RNA-seq exon coverage, aggregate (filtered), NCBI Leopardus geoffroyi Annotation Release 100 - log base 2 scaled
RNA-seq intron-spanning reads, aggregate (filtered), NCBI Leopardus geoffroyi Annotation Release 100 - log base 2 scaled


Eukaryota / Metazoa / Chordata / Mammalia / Rodentia / Caviidae / Cavia
Cavia porcellus
Domestic guinea pig (Cavia porcellus) is a species of rodent in the family Caviidae (cavies).
Browse taxonomy
Current scientific name
Cavia porcellus
Common name
domestic guinea pig, guinea pig
Taxonomic rank
species
NCBI Taxonomy ID
10141
For more details see NCBI Taxonomy
Genome
Browse all 2 genomes
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GBIF
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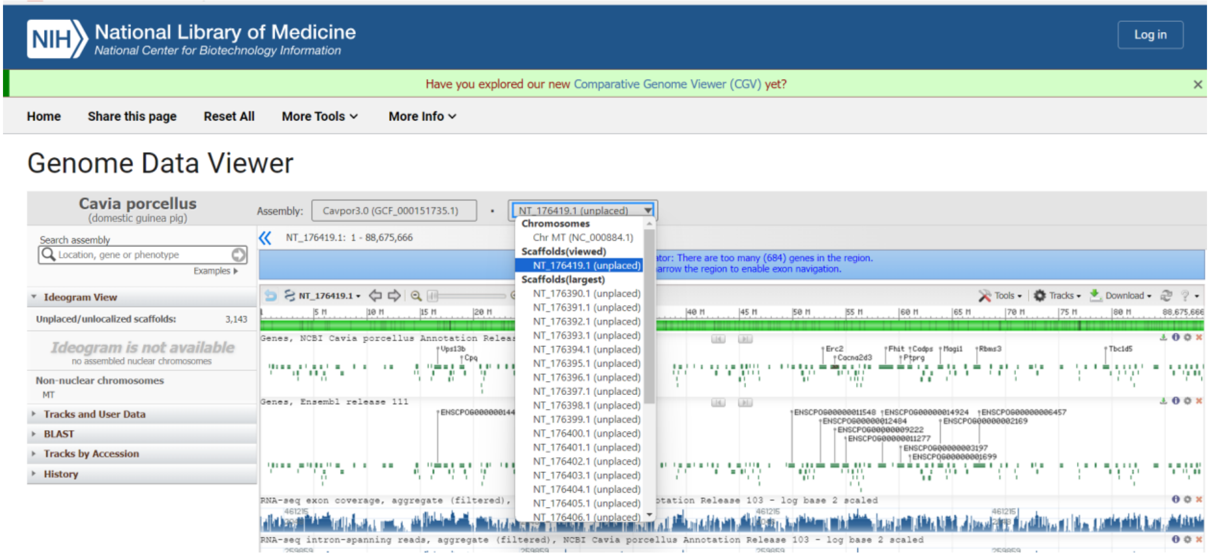
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Cavia porcellus
(domestic guinea pig)
Assembly:
Cavpor3.0 (GCF_000151735.1)
NT_176419.1 (unplaced)
NT_176419.1: 1 - 88,675,666
Search assembly
Location, gene or phenotype
Examples
Ideogram View
Unplaced/unlocalized scaffolds:
3,143
Ideogram is not available
no assembled nuclear chromosomes
Non-nuclear chromosomes
MT
Tracks and User Data
BLAST
Tracks by Accession
History
Chromosomes
Chr MT (NC_000884.1)
Scaffolds(viewed)
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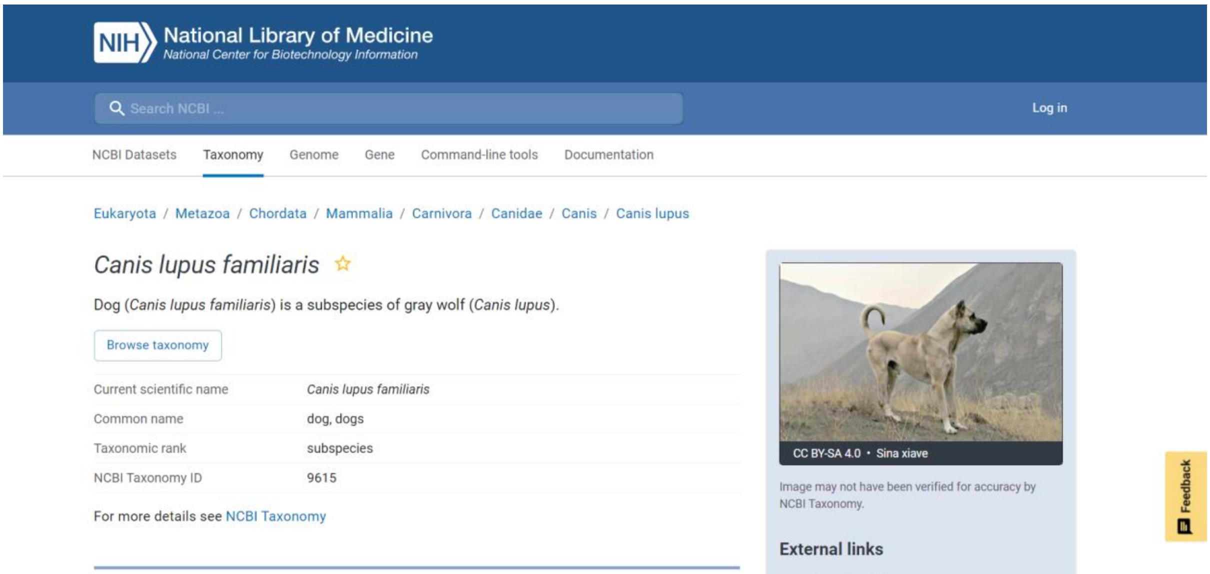
NT_176406.1 (unplaced)
Genes, NCBI Cavia porcellus Annotation Releas
Genes, Ensembl release 111
RNA-seq intron-spanning reads, aggregate (filtered), NCBI Cavia porcellus Annotation Release 103 - log base 2 scaled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Canis lupus familiaris
Dog (Canis lupus familiaris) is a subspecies of gray wolf (Canis lupus).
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Current scientific name Canis lupus familiaris
Common name dog, dogs
Taxonomic rank subspecies
NCBI Taxonomy ID 9615
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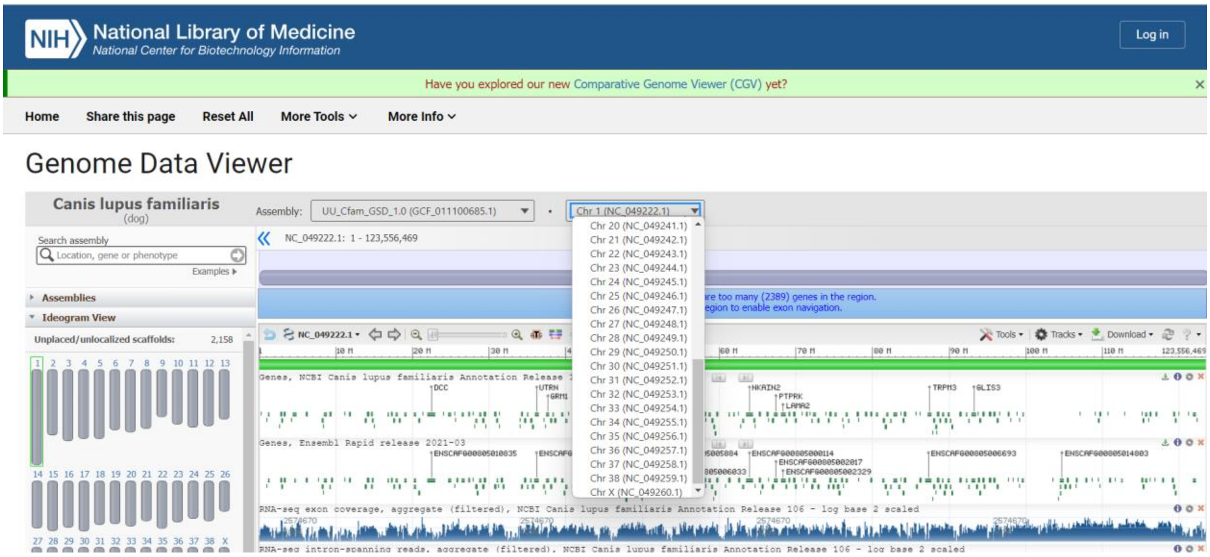
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Canis lupus familiaris
(dog)
Assembly: UU_Cfam_GSD_1.0 (GCF_011100685.1)
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Location, gene or phenotype
Examples
Assemblies
Ideogram View
Unplaced/unlocalized scaffolds: 2,158
NC_049222.1: 1 - 123,556,469
Chr 20 (NC_049241.1)
Chr 21 (NC_049242.1)
Chr 22 (NC_049243.1)
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Chr X (NC_049260.1)
Genes, NCBI Canis lupus familiaris Annotation Release
Genes, Ensembl Rapid release 2021-03
RNA-seq exon coverage, aggregate (filtered), NCBI Canis lupus familiaris Annotation Release 106 - log base 2 scaled
RNA-seq intron-spanning reads, aggregate (filtered), NCBI Canis lupus familiaris Annotation Release 106 - log base 2 scaled
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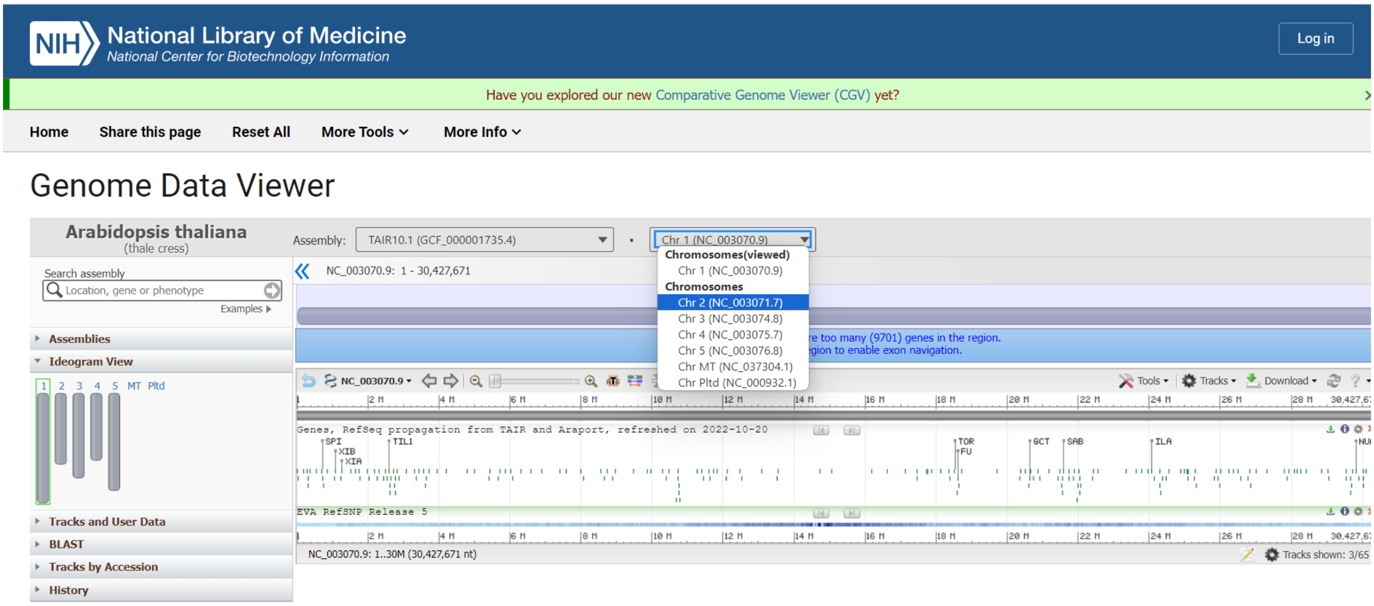
Arabidopsis thaliana
Thale cress (Arabidopsis thaliana) is a species of eudicot in the family Brassicaceae (mustard family) that is widely used as an experimental model organism.
Browse taxonomy
Current scientific name Arabidopsis thaliana
Common name thale cress, mouse-ear cress, thale-cress
Taxonomic rank species
NCBI Taxonomy ID 3702
For more details see NCBI Taxonomy

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4. **Domestic Guinea Pig:**
   - Scientific Name: Cavia porcellus
   - Taxonomy ID: 10141
   - Number of Chromosomes: 16 pairs (32 chromosomes)
5. **Thale Cress (Arabidopsis thaliana):**
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   - Taxonomy ID: 3702
   - Number of Chromosomes: 5 pairs (10 chromosomes)

10)

racemase, 5.1.1.3, P56868 advanced

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# Asparagine synthase (glutamine-hydrolysing)

Asparagine synthetase B catalyses the ATP dependent formation of asparagine from aspartate using either glutamine or ammonia as a nitrogen source. The enzyme is a homodimer with two active sites [PMID:10587437] in each subunit, one responsible for glutamine hydrolysis and one for asparagine synthesis, separated by a hydrophobic intramolecular tunnel around 20A long [PMID:12706338]. Sequencing analysis has shown the synthetase to belong to the larger glutamine transferase family.

## Reference Protein and Structure

**Sequence**

P22106 UniProt (6.3.5.4) InterPro (Sequence Homologues) (PDB Homologues)

**Biological species**

*Escherichia coli K-12* (Bacteria) UniProt

**PDB**

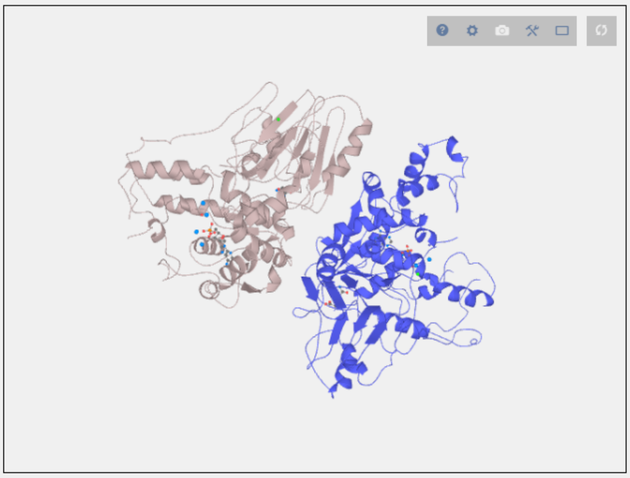
1ct9 - CRYSTAL STRUCTURE OF ASPARAGINE SYNTHETASE B FROM ESCHERICHIA COLI (2.0 Å) PDBsum RCSB PDB

**Catalytic CATH Domains**

3.60.20.10 3.40.50.620 (see all for 1ct9)

**Cofactors**

Magnesium(2+) (1) Metal MACiE



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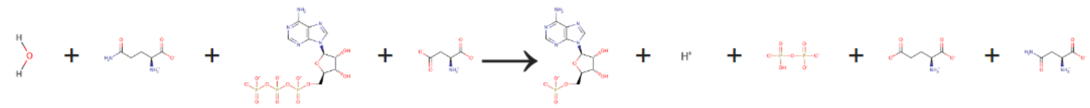
## Enzyme Reaction (EC:6.3.5.4)



| water | L-glutamine zwitterion | ATP(4-) | L-aspartate(1-) | adenosine 5'-monophosphate(2-) | hydron | diphosphate(3-) | L-glutamate(1-) | L-asparagine zwitterion |
|---|---|---|---|---|---|---|---|---|
| CHEBI:15377 | CHEBI:58359 | CHEBI:30616 | CHEBI:29991 | CHEBI:456215 | CHEBI:15378 | CHEBI:33019 | CHEBI:29985 | CHEBI:58048 |

Enzyme reaction links: IntEnz ENZYME ExplorEnz Rhea BRENDA KEGG MetaCyc

**Alternative enzyme names:** Asparagine synthetase (glutamine-hydrolyzing), Glutamine-dependent asparagine synthetase, Asparagine synthetase B, AS, AS-B,

## Enzyme Mechanism

Mechanism Proposal 1

Summary | Step 1 | Step 2 | Step 3 | Step 4 | Step 5 | Step 6 | Step 7 | Step 8 | Products | All Steps

### Introduction

The enzyme catalyses three distinct chemical reactions: glutamine hydrolysis to yield ammonia (which is then channelled to the second active site) takes place in the N-terminal domain. The C-terminal active site mediates both the synthesis of a beta-aspartyl-AMP intermediate and its subsequent reaction with ammonia. However, the exact order of the partial reactions is still somewhat unclear [PMID:9748330, PMID:12706338], here we show the hydrolysis of the glutamate occurring first.

Glutamine hydrolysis occurs at the N terminal. A nucleophilic Cys residue attacks the glutamine substrate, displacing ammonia, forming a substrate-

## Catalytic Residues Roles

| UniProt | PDB* (1ct9) | | |
|---|---|---|---|
| Cys2 (N-term) | Ala1A (N-term) | Acts as a general acid/base to activate the cysteine nucleophile. | proton acceptor, proton donor |
| Leu51 (main-C) | Leu50A (main-C) | Helps stabilise the reactive intermediates formed. | hydrogen bond acceptor, electrostatic stabiliser |
| Thr322, Arg325 | Thr321A, Arg324A | Bind and stabilise the phosphate groups of the ATP and reactive intermediates formed. | hydrogen bond donor, electrostatic stabiliser |
| Cys2 | Ala1A | Acts as a catalytic nucleophile in the glutaminase domain reaction. | covalently attached, hydrogen bond acceptor, nucleofuge, nucleophile, proton acceptor, proton donor |
| Gly76 (main-N), Asn75 | Gly75A (main-N), Asn74A | Forms the oxyanion hole that stabilises the reactive intermediates and transition states formed. | hydrogen bond donor, electrostatic stabiliser |

*PDB label guide - RESx(y)B(C) - RES: Residue Name; x: Residue ID in PDB file; y: Residue ID in PDB sequence if different from PDB file; B: PDB Chain; C: Biological Assembly Chain if different from PDB. If label is "Not Found" it means this residue is not found in the reference PDB.

## Chemical Components

proton transfer, bimolecular nucleophilic addition, proton relay, enzyme-substrate complex formation, overall reactant used, intermediate formation, unimolecular elimination by the conjugate base, enzyme-substrate complex cleavage, deamination, intermediate collapse, overall product formed, native state of enzyme regenerated

## References


1. **Tesson AR *et al.* (2003), *Arch Biochem Biophys*, 413, 23-31. Revisiting the steady state kinetic mechanism of glutamine-dependent asparagine synthetase from Escherichia coli. DOI:10.1016/s0003-9861(03)00118-8. PMID:12706338.**
2. Fresquet V *et al.* (2004), *Bioorg Chem*, **32**, 63-75. Kinetic mechanism of asparagine synthetase from Vibrio cholerae. DOI:10.1016/j.bioorg.2003.10.002. PMID:14990305.
3. Larsen TM *et al.* (2000), *Biochemistry*, **39**, 7330-. Three-dimensional structure of escherichia coli asparagine synthetase B: A short journey from substrate to product. PMID:10852734.
4. Larsen TM *et al.* (1999), *Biochemistry*, **39**, 7330-7330. Three-Dimensional Structure ofEscherichia coliAsparagine Synthetase B:  A Short Journey from Substrate to Product. DOI:10.1021/bi005109y. PMID:10587437.
5. Boehlein SK *et al.* (1998), *Biochemistry*, **37**, 13230-13238. Kinetic Mechanism ofEscherichiacoliAsparagine Synthetase B†. DOI:10.1021/bi981058h. PMID:9748330.
6. Boehlein SK *et al.* (1997), *J Biol Chem*, **272**, 12384-12392. Mutagenesis and Chemical Rescue Indicate Residues Involved in beta -Aspartyl-AMP Formation by Escherichia coli Asparagine Synthetase B. DOI:10.1074/jbc.272.19.12384. PMID:9139684.


7)

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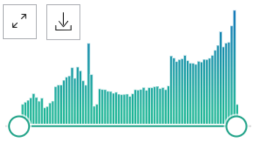


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1 Author Correction: Solving olympiad geometry without human demonstrations.
Trinh TH, Wu Y, Le QV, He H, Luong T.
**Nature**. 2024 Feb 23. doi: 10.1038/s41586-024-07115-7. Online ahead of print.
PMID: 38396303 No abstract available.
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2 Machine learning reveals huge potential benefits of sustainable fertilizer use.
[No authors listed]
**Nature**. 2024 Feb 23. doi: 10.1038/d41586-024-00431-y. Online ahead of print.
PMID: 38396101 No abstract available.
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3 The first bulk ceramic that deforms like a metal at room temperature.
[No authors listed]
**Nature**. 2024 Feb 23. doi: 10.1038/d41586-024-00443-8. Online ahead of print.
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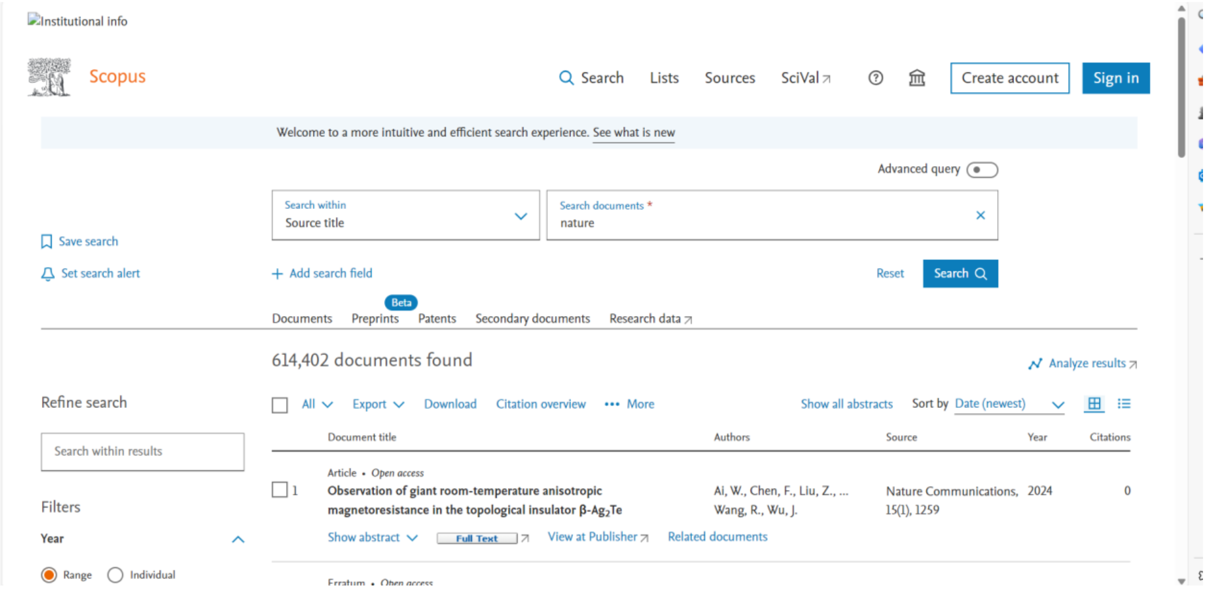
| 1 | PMID | Title | Authors | Citation | First Authc | Journal/Bc | Publicatior | Create Dat | PMCID | NIHMS ID | DOI | | |
| 2 | 38396303 | Author Coı | Trinh TH, V | Nature. 20 | Trinh TH | Nature | 2024 | ######## | | | 10.1038/s41586-024-07115-7 | | |
| 3 | 38396101 | Machine learning reve | | Nature. 2024 Feb 23. | | Nature | 2024 | ######## | | | 10.1038/d41586-024-00431-y | | |
| 4 | 38396100 | The first bulk ceramic | | Nature. 2024 Feb 23. | | Nature | 2024 | ######## | | | 10.1038/d41586-024-00443-8 | | |
| 5 | 38396099 | 'All of Us' ı | Kozlov M. | Nature. 20 | Kozlov M | Nature | 2024 | ######## | | | 10.1038/d41586-024-00568-w | | |
| 6 | 38396098 | Neuralink I | Drew L. | Nature. 20 | Drew L | Nature | 2024 | ######## | | | 10.1038/d41586-024-00550-6 | | |
| 7 | 38396097 | Daily briefi | Graham F. | Nature. 20 | Graham F | Nature | 2024 | ######## | | | 10.1038/d41586-024-00571-1 | | |
| 8 | 38396096 | How we bı | Byrne D. | Nature. 20 | Byrne D | Nature | 2024 | ######## | | | 10.1038/d41586-024-00212-7 | | |
| 9 | 38396095 | What's the | Padamana | Nature. 20 | Padamana | Nature | 2024 | ######## | | | 10.1038/d41586-024-00572-0 | | |
| 10 | 38396094 | CAR-T theı | Mullard A. | Nature. 20 | Mullard A | Nature | 2024 | ######## | | | 10.1038/d41586-024-00470-5 | | |
| 11 | 38396093 | First privat | Witze A. | Nature. 20 | Witze A | Nature | 2024 | ######## | | | 10.1038/d41586-024-00549-z | | |




12)

```
In [2]: import requests

def fetch_fasta_record(db, accession):
    base_url = "https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi"
    params = {
        'db': db,
        'rettype': 'fasta',
        'id': accession,
    }

    response = requests.get(base_url, params=params)

    if response.status_code == 200:
        return response.text
    else:
        return f"Error: {response.status_code} - {response.text}"

# Example usage:
database_name = "nucleotide"
accession_number = "NM_000520"
fasta_sequence = fetch_fasta_record(database_name, accession_number)

print(fasta_sequence)
```

```
>NM_000520.6 Homo sapiens hexosaminidase subunit alpha (HEXA), transcript variant 2, mRNA
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TGCTGCTGGCGGCAGCGTTCGCAGGACGGGCGACGGCCCTCTGGCCCTGGCCTCAGAACTTCCAAACCTC
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13)

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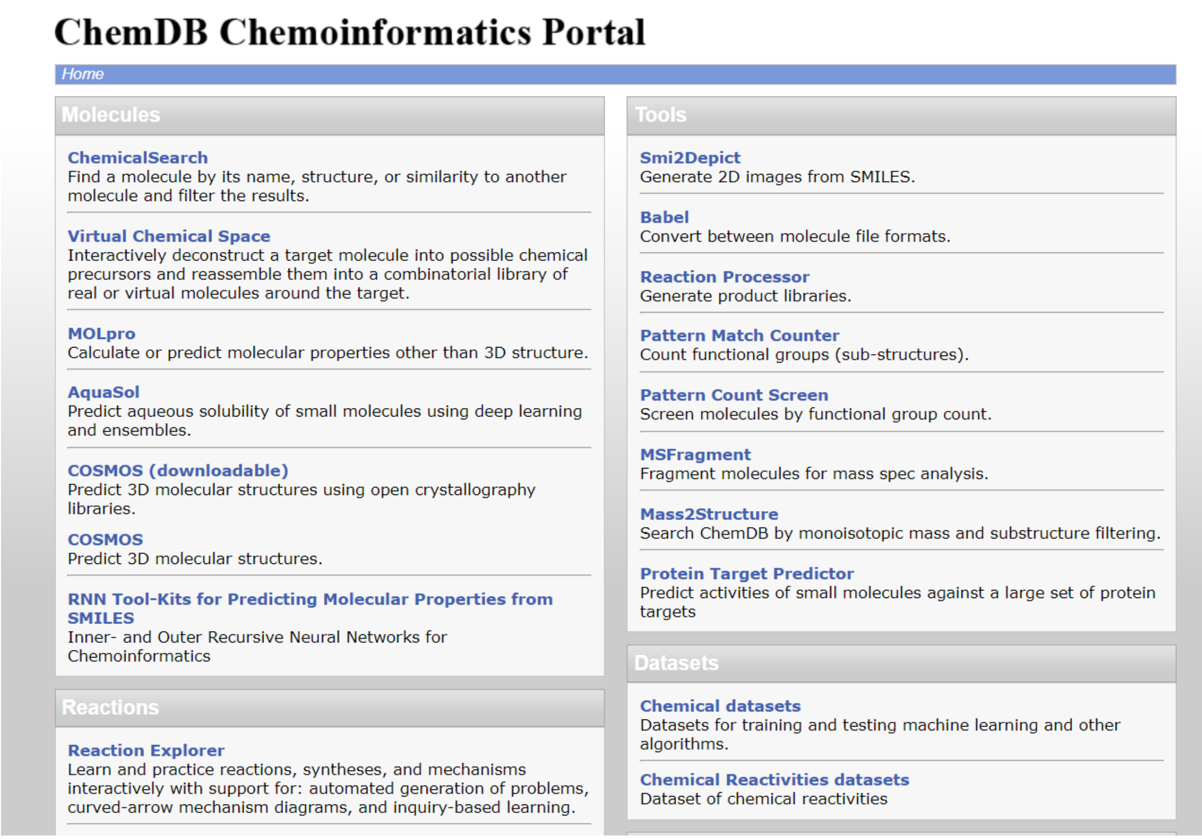


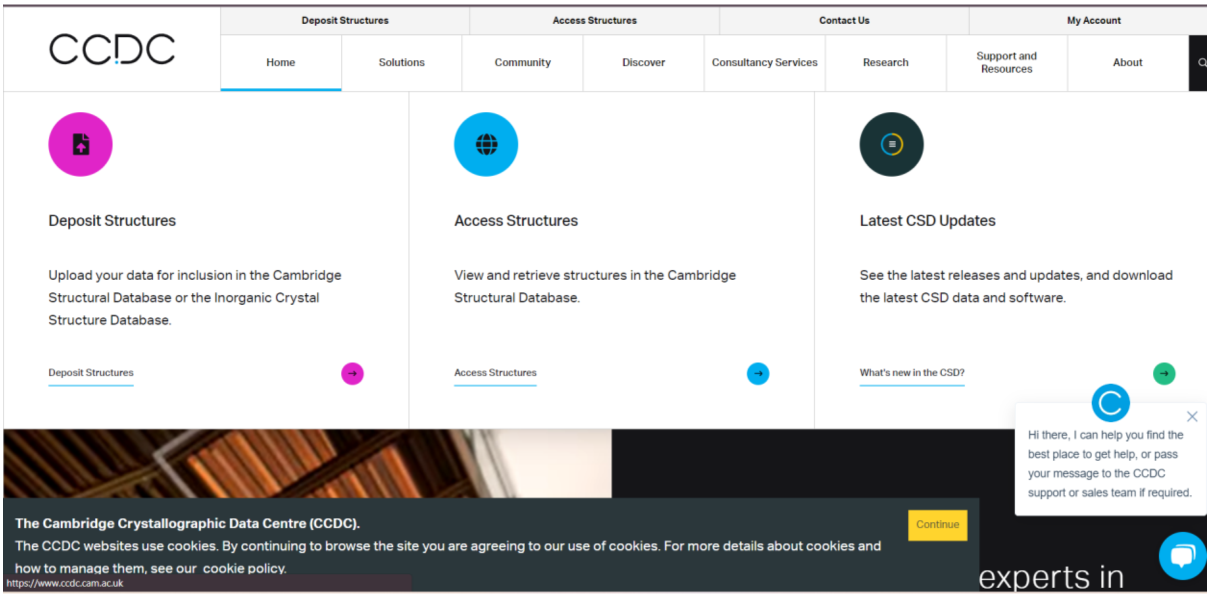


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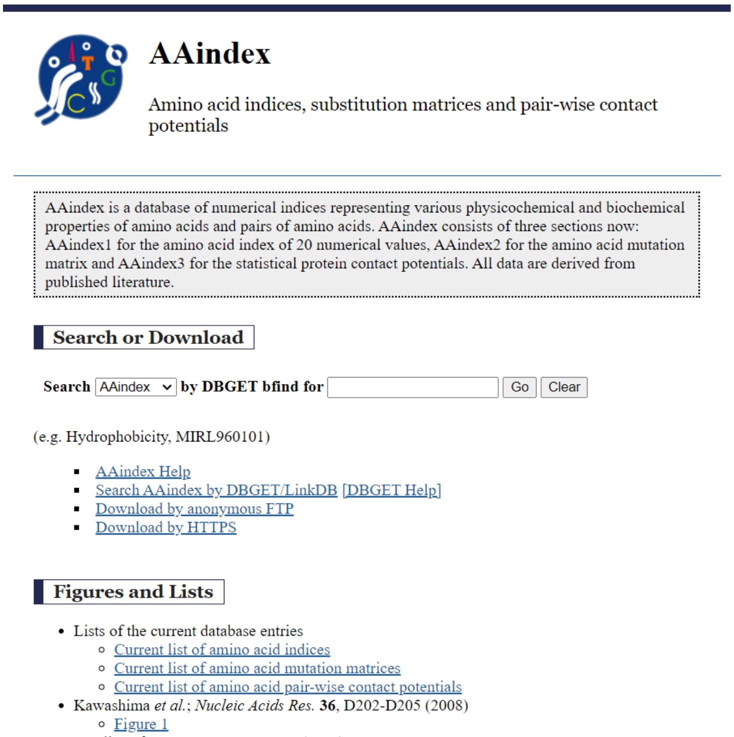


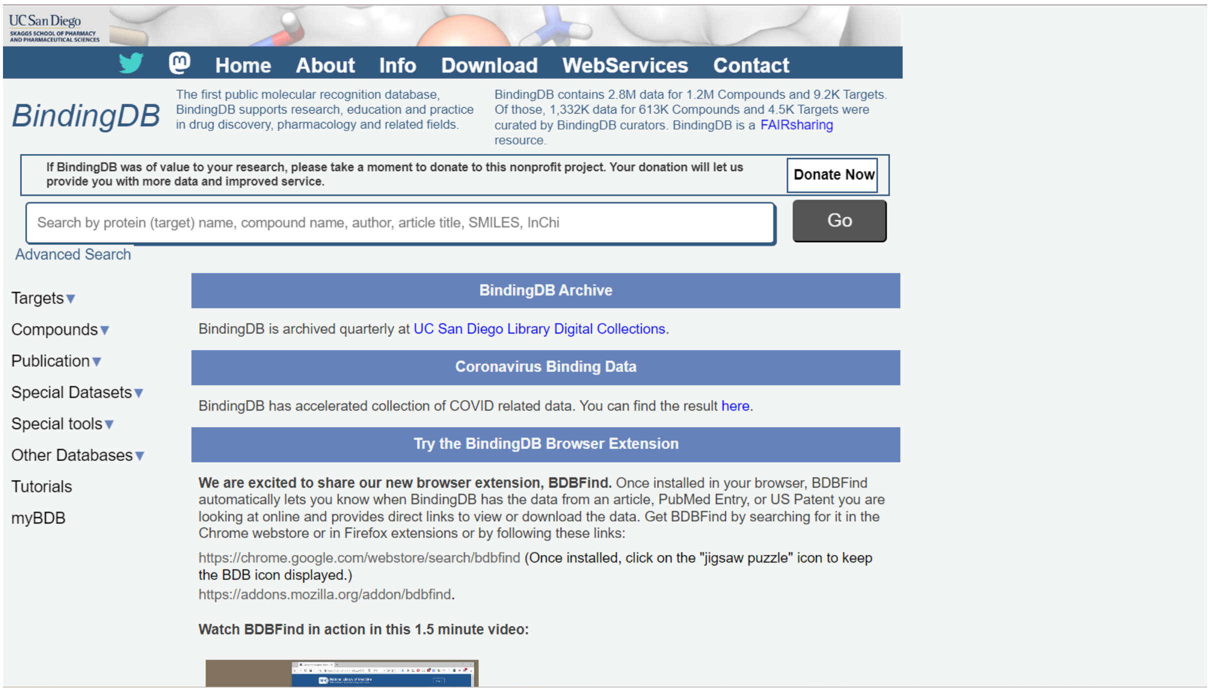


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  - Candidate Cancer Gene Database
  - CanGEM
  - CanSAR
  - CaSNP
  - cBioPortal
  - CCDB
  - ccmGDB
  - CellLineNavigator
  - ChimerDB
  - CIViCdb
  - ClinVar
  - CMPD
  - Colorectal Cancer Atlas
  - COLT-Cancer
  - COSMIC
  - CTDatabase
  - dbDEMC
  - dbDEPC
  - DIVAS
  - DriverDBv2
  - FusionCancer
  - HLungDB
  - HPTAA
  - IARC TP53 Database
  - IGDB.NSCLC
  - intOGen
  - ITTACA
  - lnc2cancer
  - MethHC
  - MethyCancer
  - MoKCa
  - MutationAligner
  - Network of Cancer Genes